

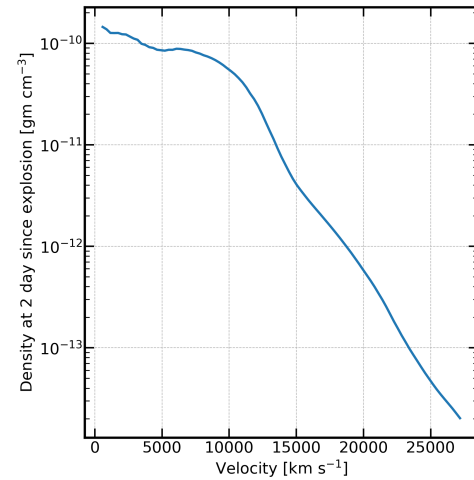


Understanding Type Ia Supernova explosions through time dependent Monte Carlo radiative transfer simulations

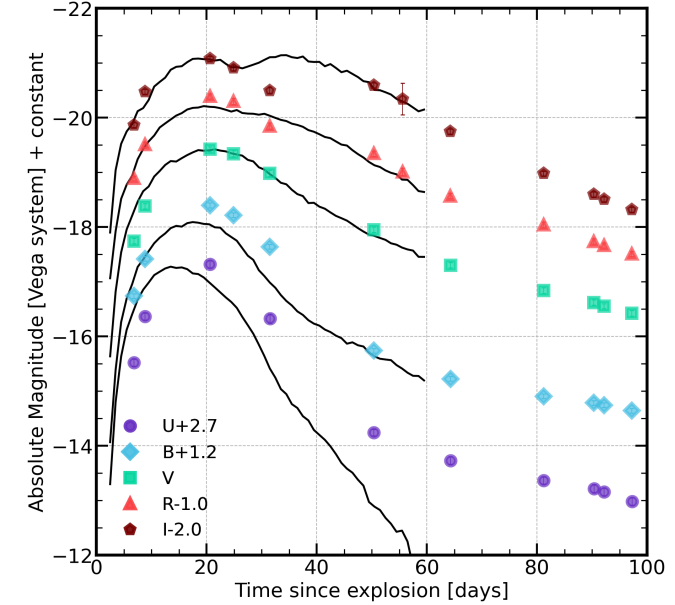
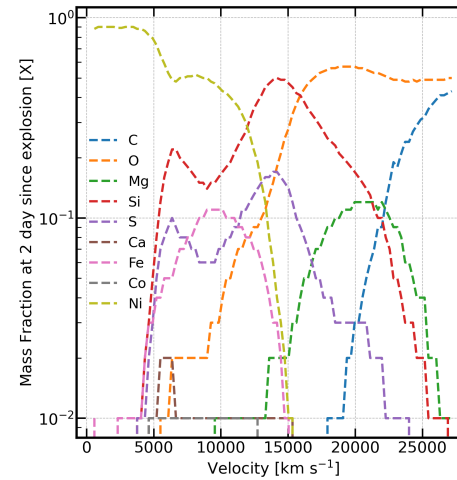


Anirban Dutta¹, G C Anupama¹, D K Sahu¹ and Varun Bhalerao²

Abstract : We present light curves of a normal type Ia supernova SN 2019np observed with the 2.0 m Himalayan Chandra Telescope. We simulate light curves with Monte Carlo radiative transfer code SEDONA to explore the explosion parameters of SN 2019np. We also simulate light curves to understand the distribution of the synthesised ^{56}Ni in the ejecta of type Ia supernova from a $1.4 M_{\text{ch}}$ white dwarf assuming a deflagration-to-detonation density profile.



Density profile of the DDT explosion model N100 (Sim+ 2013).
Abundance profile of the elements in the ejecta of SN 2019np.



Sedona is a multi-dimensional Monte Carlo radiative transfer code for simulating Ni-powered transients. (Kasen, Thomas and Nugent, 2006)

Assumptions: (i) Homologous Expansion
(ii) Sobolev Approximation (iii) LTE

Three dimensional hydrodynamical simulation of the explosion of a $1.4 M_{\odot}$ WD initially composed of 47.5% ^{12}C , 50% ^{16}O and 2.5% ^{22}Ne by mass. The white dwarf central density is $2.9 \times 10^9 \text{ g/cm}^3$. The explosion consists of 100 ignition points placed randomly around the centre of the white dwarf. (Seitenzahl+ 2013, Sim+ 2013).

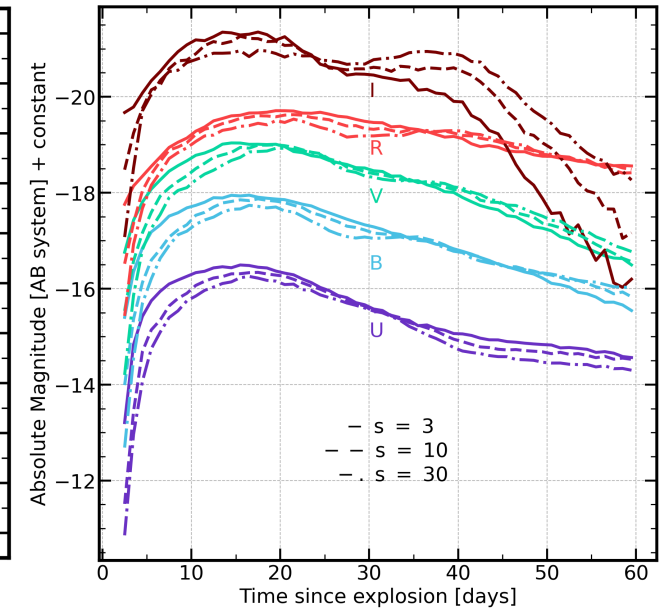
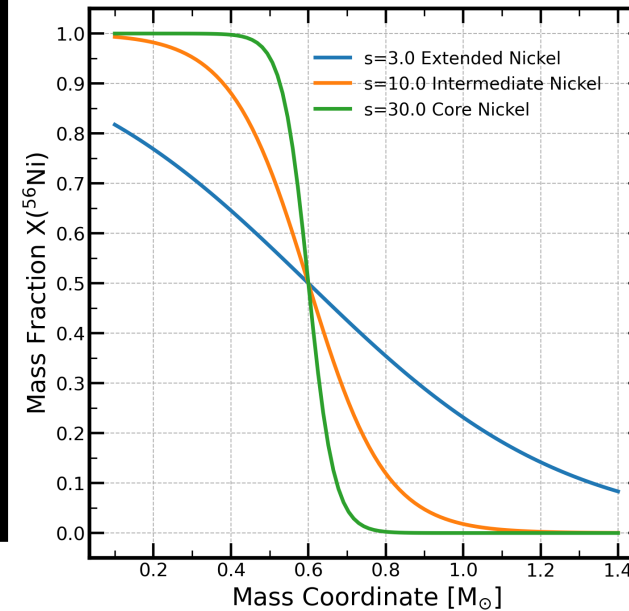
$$X(^{56}\text{Ni}) = \frac{1}{\exp(s(m - M_{\text{Ni}})/M_{\odot}) + 1}$$

$$X(^{16}\text{O}) = \frac{1}{\exp(-s(m - M_{\text{Ni}})/M_{\odot}) + 1}$$

$$X(^{32}\text{Si}) = 1 - X(^{56}\text{Ni}) - X(^{16}\text{O})$$

The parameter 's' controls the shape of the Ni-distribution.

Explosion parameter of SN 2019np: ^{56}Ni mass = $0.65 M_{\odot}$, Kinetic energy : $1.47 \times 10^{51} \text{ erg}$.



- With extended ^{56}Ni ($s=3$), we find that the light curve is bright in the earliest phase.
- The secondary maximum in near-IR (I-band) is not prominent in a fully mixed profile.
- In the core-Ni model ($s=30$), the light curves show distinct secondary maximum.
- The $s=30$ model shows that the U-flux is distributed to near-IR.